Lesson 4: Ramp Practical and Line Graphs

Unit

Working Like a Physicist: Collecting, Processing, and Presenting Data

Lesson focus

Students collect continuous data, record repeated measurements, calculate a mean, and begin plotting a line graph on graph paper.

Assumptions

Students have already learned that continuous independent variables usually require line graphs.

Practical title

How does ramp height affect the distance travelled by a trolley?

Investigation question

How does the height of a ramp affect the distance travelled by a trolley after leaving the ramp?

Why this practical works well

This practical is active, simple, and engaging. It gives continuous numerical data and leads naturally to a line graph.

Learning objectives

By the end of the lesson, students should be able to:

1. Identify continuous independent and dependent variables.
2. Draw a results table with 3 trials and a mean.
3. Collect data efficiently within a time limit.
4. Decide that a line graph is appropriate.
5. Plot points accurately on graph paper.
6. Label graph axes with quantities and units.

Key vocabulary

Keyword	Student-friendly definition
Continuous data	Number data that can have many possible values
Independent variable	The thing you change
Dependent variable	The thing you measure
Control variable	Something you keep the same
Mean	The average value
Line graph	A graph used when the independent variable is continuous

Equipment

Per group:

- Ramp
- Trolley or toy car

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- Meter ruler or tape measure
- Blocks or books to change ramp height
- Graph paper
- Exercise book or results table sheet

Safety notes

- Keep bags and stools away from the testing area.
- Do not launch trolleys toward people.
- Students should collect trolleys carefully after each run.
- Make sure ramps are stable before releasing the trolley.

Starter

Ask:

> *Ramp height is measured in cm. Is this continuous or categoric?*

Expected answer:

Ramp height is continuous.

Then ask:

> *What graph should we use?*

Expected answer:

A line graph.

Variables

Variable type	Example
Independent variable	Ramp height in cm
Dependent variable	Distance travelled in cm
Control variables	Same trolley, same ramp, same starting point, same surface

Method

Students use 5 different ramp heights, for example:

- 5 cm
- 10 cm
- 15 cm
- 20 cm
- 25 cm

For each height:

1. Set the ramp to the chosen height.
2. Place the trolley at the same starting point on the ramp.
3. Release the trolley without pushing it.

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4. Measure the distance travelled after the trolley leaves the ramp.
5. Repeat 3 times.
6. Calculate the mean distance.
7. Repeat for the other ramp heights.

Example results table

Ramp height / cm	Distance trial 1 / cm	Distance trial 2 / cm	Distance trial 3 / cm	Mean distance / cm
5				
10				
15				
20				
25				

Data processing

Students begin plotting a line graph.

Graph setup:

- x-axis: ramp height / cm
- y-axis: mean distance / cm

Graphing expectations

A good line graph should have:

1. A clear title.
2. Independent variable on the x-axis.
3. Dependent variable on the y-axis.
4. Units on both axes.
5. A sensible scale.
6. Accurately plotted points.
7. Small crosses, not large dots.
8. A best-fit line, not a dot-to-dot line.

In this lesson, the priority is accurate setup and plotting. Best-fit lines can be developed more fully in Lesson 5.

Common graphing mistakes to watch for

- Missing units on axes.
- Poor scale choices, such as awkward intervals.
- Starting the graph too small when the whole page could be used.
- Putting the dependent variable on the x-axis.
- Drawing bars instead of a line graph.
- Joining point to point before discussing best-fit lines.

Discussion questions

1. Why is ramp height a continuous variable?

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2. Why is a line graph suitable here?
3. Why should we calculate a mean?
4. What could cause random error in this experiment?
5. What control variables were difficult to keep the same?

Plenary

Students answer:

1. The independent variable was _____.
2. The dependent variable was _____.
3. A line graph was suitable because _____.
4. One control variable was _____.

Possible prep

Students finish plotting the graph if needed. They should check that their graph includes a title, labeled axes, units, sensible scale, and accurately plotted points.

Teacher notes

The practical may produce messy data, especially if the trolley veers sideways or if the floor surface changes. This is useful for discussing random error and the need for repeats.

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